

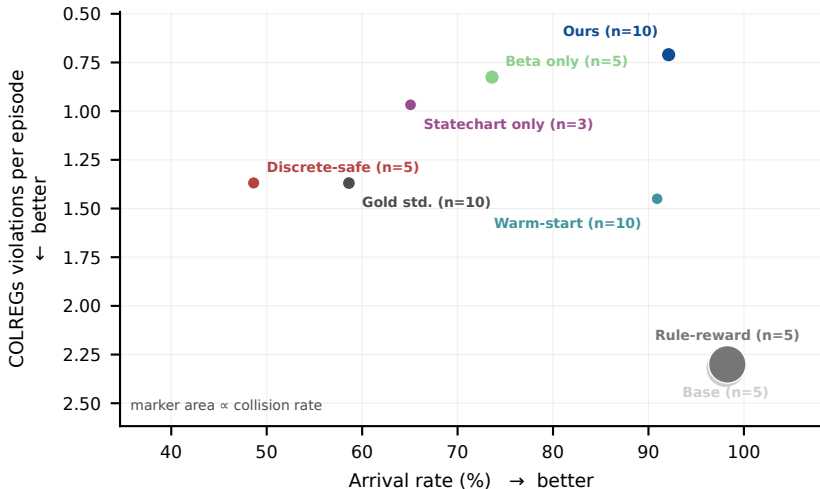


Fig. 3 | Safety-efficiency trade-off on the official test set. Each marker is one arm (mean over its seeds); marker area is proportional to the pooled collision rate. Up-and-left is better. Arms: Base and Rule-reward are discrete without a shield; Discrete-safe is the discrete action-masking benchmark; Gold standard and Warm-start are the previous continuous recipe; Beta only and Statechart only are single-mechanism ablations. 5.08M-step budget · last checkpoint · official test set (strict 563) · single machine, single pass